Detection of Payload-Free Pretexting Emails Through Claim-Routed Verification Against Email Headers and Thread History

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***Abstract*—Pretexting emails often carry no link and no attachment, and a sender who writes from a free mailbox or a look-alike domain passes SPF, DKIM and DMARC for that domain. This paper describes a claim-routed verification pipeline that extracts typed claims from the body and routes each one to the header or thread rules that can test it. Every verdict is a ledger row naming the claim, the rule and the evidence, so each of the 442 test emails the pipeline flagged has a stated reason. On synthetic takeover replies injected into real threads, the thread rules detect 18 of 25 Apache and 48 of 51 Enron cases, against 5 and 0 for single-message rules. A payload-free protocol, run on corpus labels with one training sample and two seeds, shows that part of a raw-trained detector's accuracy is link-reading: its F1 drop under link redaction exceeds that of a redaction-trained model by 0.073, with a bootstrap interval from 0.061 to 0.086. The pipeline is not the best pooled detector. On 3,045 test attacks and 6,357 legitimate emails its F1 is 0.238, against 0.882 for parallel fusion of text and header classifiers, and its AUC is the lowest of five systems.**

Keywords—business email compromise, pretexting, phishing detection, email authentication, claim verification, thread hijacking

I. Introduction

Business email compromise (BEC) is pretexting aimed at money: the attacker poses as a trusted person, such as an executive or a supplier, and asks for a payment, a change of bank details or data. In 2025 the Internet Crime Complaint Center of the FBI recorded 24,768 BEC complaints and $3,046,598,558 in reported losses [1]. The 2026 Data Breach Investigations Report defines pretexting as building a trusted relationship through invented scenarios, by voice, email or text, with the attacker taking part in the conversation or email thread, and finds it to be the initial access vector in 6% of breaches [2].

Such emails can be payload-free: they carry no link and no attachment, so scanners built around URLs and attachments find nothing. Consider a message signed by David from Finance that asks an accounts clerk at acmecorp.com to process a wire transfer before 3 PM and to tell no one. It comes from a real gmail.com account, so SPF [3], DKIM [4] and DMARC [5] all pass, for gmail.com. Authentication proves which domain sent a message; it says nothing about whether the sender is the person or the organisation the body claims. The attack is visible only as a disagreement between what the body claims and what the headers prove.

Detectors that could see this disagreement mostly look elsewhere. Work on phishing from compromised accounts focuses on emails with a malicious URL, and its authors note that their data likely underestimate attacks without one [6]. BEC-Guard judges impersonation from an organisation's header history and classifies the body only after removing names and signatures [7], so the statements that say who is writing are never compared with the headers. ChatSpamDetector asks GPT-4 to look for brand impersonation and header spoofing, which covers part of this comparison, but its data exclude emails without a URL and each email costs one model call [8].

We treat the problem as fact-checking. Automated fact-checking divides the task into claim detection, evidence retrieval and claim verification, which gives a verdict and its justification [9]. Here the claims are what an email says about itself and the evidence is its own headers and, for a reply, the earlier messages of its thread. A claim to be a colleague is checked against the authenticated domain, a payment request against who is asking, and a reply against the history it joins. A taken-over account passes every header check, so for that attacker the thread is the only evidence left.

The paper makes four contributions. 1) The claim-routed verification pipeline, in which typed claims are routed to the header, request or thread rules that can test them and every finding is a ledger row naming the claim, the rule and the evidence. 2) A payload-free evaluation protocol, N1, which redacts links, addresses, domains and file names and measures how much of a detector's accuracy is link-reading. 3) Thread consistency verification, N2, which compares a reply with its own thread. 4) An evaluation with frozen components, a test log, grouped bootstrap intervals and per-source reporting, which also shows that the pipeline is not the best pooled detector on public corpora.

II. Related Work

Persuasion and tactics. Persuasion is measurable in email text. Karki et al. tag persuasion principles in phishing emails with transformer classifiers and report an average F1 of 0.9 for RoBERTa on five principles, with labels produced automatically by sentence similarity to hand-written examples [10]. Tooher and Lallie report 94% accuracy for a two-stage persuasion-based detector, but their phishing emails were generated by a language model and labelled by design while their legitimate emails are real [11]; in such a set, writing style alone can separate the classes. Across seven public corpora, Pan et al. find every persuasion principle more often in phishing than in legitimate mail, yet authority cues still appear in 27.10% of legitimate samples, and the authors note that targeted attacks such as BEC may not be represented in such corpora [12]. These studies read the body only. In the pipeline a tactic is a modifier: it raises the weight of a contradiction and never creates one.

Identity from metadata and history. BEC-Guard first decides from an organisation's header history whether a sender impersonates an employee and only then classifies the body and its links, after removing names and signatures from the text; it reports a precision of 98.2% on more than 4,000 attacks, and account takeover is outside its focus [7]. Gascon et

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al. profile senders from the structure and transport traits of their emails without reading the text, detecting 90% of spoofed emails at fewer than one false alarm in 10,000; they note that such infrastructure features model the sender's environment rather than the author, so a hacked account cannot be detected with them [13]. These systems ask whether an email fits its sender's history. The pipeline asks whether what the email claims matches its own headers, needs no mail history beyond the thread being checked, and meets the taken-over account in the thread. We did not find an academic detector of thread hijacking with an open evaluation.

Claims and model judgement. Paul et al. annotate 940 emails with the warning signs that users are trained to notice, among them the organisation a sender claims, and train a classifier for each sign; they observe that a mismatch between such a claim and the metadata is often a stronger signal than the content, and assume a downstream verification algorithm for that comparison [14]. The pipeline is such an algorithm. ChatSpamDetector reports 99.70% accuracy on 2,010 emails with a GPT-4 prompt that includes a check for a mismatch between the sender name and address [8], so this paper does not claim to be the first to compare a claimed sender with the headers. Conversation-level detection has been studied for chat: ConvoSentinel classifies social-engineering attempts at message and conversation level in 1,400 generated conversations [15], without the header and thread metadata that email provides. The pipeline differs in checking typed claims with fixed rules that record their reasons, with no model call at run time, and in being evaluated on link-free, redacted text.

III. Problem and Threat Model

An email is payload-free when it carries no link and no attachment, so the attack lies entirely in what the text asks the reader to do. The defender is the receiving organisation. The input is one email or a thread, with the organisation's own domain when it is known; when it is not, checks that need it are recorded as not checkable rather than guessed.

Outside impersonator. The attacker writes from a free mailbox, a look-alike domain or a spoof of the organisation's own domain, often with a display name and a Reply-To chosen to mislead. A free-mailbox or look-alike sender passes SPF, DKIM and DMARC for its own domain; an exact-domain spoof fails DMARC when the organisation publishes a policy. In both cases the headers do not support the affiliation the body claims, which is what the header rules (N3) look for.

Attacker inside a real account. After an account takeover, a vendor compromise or a thread hijack, mail comes from the real sender's domain and passes authentication, so the header rules have nothing to contradict. What changes is the conversation: new urgency, new bank details, a new sending path. The thread rules (N2) look for that, and they also test a forged thread, a new email that quotes a conversation that never happened.

Trust in headers. Every header is written by the sender except those the receiving side adds. Only Authentication-Results fields added by the receiving organisation are trusted, as RFC 8601 requires [16]; the pipeline reads their verdicts and does not recompute them, and a missing verdict is unknown, never a pass. Out of scope is an attacker who has read the mailbox and copies the reply identifiers, the sending details and the calm tone of the real sender; no rule here can catch that message.

IV. The Claim-Routed Verification Pipeline

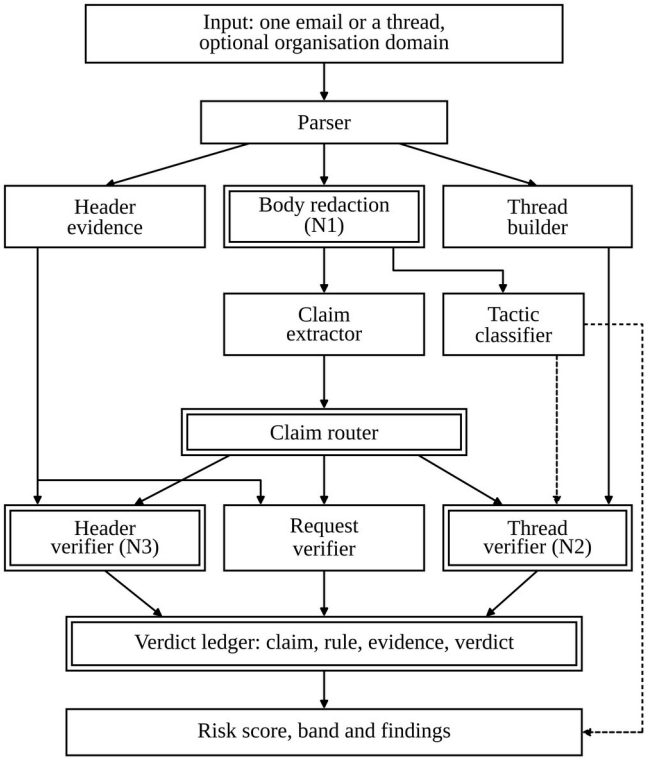


Fig. 1. The claim-routed verification pipeline. Double borders mark the parts introduced here; the dashed line carries tactic probabilities to the thread verifier (tactic onset) and to the score (urgency and secrecy as modifiers).

Fig. 1 shows the pipeline. A parser splits each message into headers, body and thread links, and three branches follow: header evidence, body redaction and, for a thread, a thread builder that orders the messages and keeps each one's quoted history.

A. Redaction, Claims and Tactics

The cleaned body is redacted before any model or rule reads it: URLs, e-mail addresses, bare domains and attachment file names become the placeholders [URL], [EMAIL], [DOMAIN] and [FILE], while words such as *see attached* stay. A claim extractor built from a tokenizer, a name recogniser and 85 token patterns reads the redacted text and its signature block; nothing in it is trained. It emits typed claims with a text span: internal or external affiliation, authority, reply direction and signature contact; prior relationship; and five request types (payment, payment change, credential, gift card, data). Beside the extractor, a tactic classifier gives probabilities for seven tactics: authority, scarcity, reciprocity, social proof and liking from the persuasion literature [17], urgency from the time principle [18], and isolation or secrecy.

B. Routing and Verification

The router sends affiliation, authority, reply-direction and signature claims to the header verifier, a prior-relationship claim to the thread verifier, and each request to the request verifier and, when a thread is present, also to the thread verifier. Tactics go to no verifier, because no header can prove or disprove urgency.

The header verifier (N3) compares three domains: the From domain the reader sees, the domain that authentication vouched for and the domain the claim implies, which is the organisation's own domain or a known brand's domains. A contradiction is a mismatch that the claim makes meaningful, so the same header values mean different things under different claims: dmarc=pass for gmail.com is normal for a

private Gmail user and a high-severity contradiction under *this is David from Finance*. The request verifier asks who is asking: a free mailbox outside the organisation, a look-alike sender, a failed DMARC check or a Reply-To to another domain. A change of bank details can never be confirmed by headers and is left to the thread.

The thread verifier (N2) judges each message against the messages before it on four signals: tactic onset, where a tactic crosses its threshold for the first time and well above its earlier average; request drift, such as new bank details or a first credential request; sending-path drift, such as the same address from a new network and mail program or the sender's name at a look-alike domain; and thread integrity, where reply identifiers point to no earlier message or quoted text never appeared in the thread. The first message with a medium or high contradiction is reported as the point where the thread flipped.

C. Verdict Ledger and Risk Score

Every verifier writes ledger rows with one of three verdicts: contradiction with a severity of high, medium or low; consistent; or not checkable when the evidence is missing. Each row names the claim, the rule, the evidence and a reason built from fixed templates. The risk score uses contradictions only. A high row is worth 60 points, a medium 35 and a low 5; the strongest row of a claim counts, and each further claim counts half as much as the one before. Three rules that fire often on legitimate mail count zero. Urgency and secrecy each raise the multiplier on these points by 0.25, and the four main tactics add 4 points each when they fire, at most 12, which cannot reach the Suspicious band at 35 on their own; High risk starts at 70. The weights were fixed on validation data before the test split was read. Table I traces the example of Section I.

TABLE I. LEDGER FOR THE EXAMPLE OF SECTION I

Claim and span	Verifier	Evidence	Row
affiliation_internal “this is David from Finance”	header	From gmail.com, a free mailbox; authenticated as gmail.com; organisation acmecorp.com	contradiction, high hv_int_freemail 60 points
payment_request “process a wire”	request	payment request from a free mailbox outside acmecorp.com	contradiction, high rv_freemail +30 points

Score 90, High risk, before any tactic points. The Reply-To to protonmail.com appears in the findings but adds no row, because the body makes no reply-direction claim.

The pipeline is written in Python. The tactic classifier is DistilBERT [19] fine-tuned on the labelled real and synthetic train emails of Section V, with thresholds tuned on validation. No language model runs during analysis, and an analysed email is held in memory only for the request.

V. EVALUATION DESIGN

The evaluation follows the pitfalls catalogued by Arp et al. [20]: the test split was frozen before it was read, results are reported per source, and a style-confound test measures how far the collections differ by wording alone.

A. Data and Labels

Nine public sources give 99,324 unique emails, 20,313 of them attacks: phishing from the Nazario corpus [21] and from honeypot collections [22], fraud, legitimate mail and spam from the CEAS-08, Enron [23], Ling and Nigerian Fraud files of a merged corpus [24], the SpamAssassin public corpus [25], and two Apache user lists [26] as modern legitimate mail with authentication headers. The split is 70/15/15 by source and category, with emails of one subject kept together: 69,542 train, 14,879 validation and 14,903 test emails. No source holds both attacks and legitimate mail, so a pooled score mixes corpus and detection; results are also given per source.

Real threads were rebuilt from the raw Enron corpus by subject, date and participants and from the Apache lists by reply identifiers: 1,591 threads, split by thread. In each test base thread a synthetic message replaces the next real reply in one of three variants: A, an account takeover whose text drifts while the sender, server, mail program and reply identifiers are copied from the real sender; B, a look-alike domain swap with benign text and a new sending path (Apache only, as raw Enron has no sending-path headers); C, a forged thread with a fabricated quotation and, on Apache, unknown reply identifiers. The negatives are the real next reply and a synthetic benign continuation. The injected texts were written by one language-model family.

For the tactics and claims, 700 real emails were drawn, a fixed number per source and category. Two language-model annotators of one model family labelled seven tactics and eleven claim types, and a third model of the same family broke ties; 690 emails were labelled, 140 of them in the test split. Mean Cohen's kappa between the two annotators was 0.501 over the tactics and 0.458 over the claim types. No human labels were collected, although LLM labels need validation against human labels task by task [27], so every tactic or claim score is agreement with one model family. 222 synthetic attack-and-twin pairs cover rare tactics and are reported apart.

B. Experiments

Payload-free protocol (N1). Two DistilBERT models separate attacks (phishing, fraud) from other mail (ham, spam): model A is trained on raw bodies and model B on redacted bodies, from one training sample of 9,856 emails with two seeds and a fixed threshold of 0.5. A TF-IDF and logistic regression pair, fitted on the same sample and on all train emails, is a second family. Each model is tested on the raw and redacted views of the test split and on its naturally link-free emails. The quantity of interest is A's drop in attack-class F1 from raw to redacted minus B's drop: a model that reads links as a shortcut [28] loses more when they are removed. The labels are corpus labels.

Thread consistency (N2). The thread verifier is compared with the header and request verifiers alone on the test cases; a case is detected when the injected message receives a medium or high contradiction.

Claim conditioning (N3) and architecture. The frozen routed score, flagging at Suspicious or above, is compared with four logistic regressions fitted on the train split, attack against ham: text only (tactic probabilities and claim counts), headers only (authentication verdicts, free mailbox, look-alike, Reply-To and similar fields), parallel fusion (the mean of the text and headers scores) and a flat classifier given every ledger signal as one vector without the link between a claim and its evidence. Each learned system is held to the false-alarm rate the frozen score shows on validation ham, 0.14%, with a floor of 0.5%. A synthetic colleague-impersonation arm adds 31 attack-and-twin pairs whose header blocks were written by rule: the attack comes from a free mailbox or a look-alike domain that passes its own authentication, the twin from the organisation's own domain.

C. Protocol

Before the first test run a freeze record stored a SHA-256 hash of every frozen file (the patterns, the rules, the score weights, the model weights and the thresholds), and every script refused to read the test split if one differed. Each script

was rehearsed on validation and read the test split once. Four evaluation scripts were later rerun once after two reporting bugs were fixed (per-source rows that mixed ham and spam, and style-confound pairs that mixed kinds); the log records each rerun with its reason, and no number reported here changed. Intervals are 95% percentile intervals from 1,000 bootstrap resamples that draw whole groups (subject groups for emails, threads for hijack cases, pairs for synthetic emails), and differences between systems are paired. Cells with fewer than 10 positives are reported as counts.

VI. Results

TABLE II. Main Results on the Test Split

(a) Pooled detection: 3,045 attacks against 6,357 ham

System	F1 [95% interval]	Recall	FAR	AUC
Routed score	0.238 [0.204, 0.277]	13.6%	0.3%	0.812
Text only	0.639 [0.612, 0.666]	47.4%	0.5%	0.949
Headers only	0.788 [0.767, 0.810]	65.5%	0.3%	0.966
Parallel fusion	0.882 [0.869, 0.895]	80.3%	0.8%	0.993
Flat classifier	0.658 [0.629, 0.687]	49.7%	0.5%	0.949

(b) Synthetic messages in real threads: cases flagged

Case	Source	Cases	Thread rules	Single-message rules
A, takeover	Apache	25	18	5
A, takeover	Enron	51	48	0
B, look-alike swap	Apache	17	17	0
C, forged thread	Apache	25	25	0
C, forged thread	Enron	51	45	0
Real next reply	Apache	25	0	0
Real next reply	Enron	51	1	0

(c) Links: DistilBERT attack-class F1; corpus labels; 9,856 training emails, two seeds

Model	Raw	Redacted	Raw minus redacted
A, trained on raw	0.935	0.877	+0.059 [+0.050, +0.069]
B, trained on redacted	0.918	0.932	−0.014 [−0.019, −0.009]
A's drop minus B's drop			+0.073 [+0.061, +0.086]
Same, TF-IDF, sample			+0.035 [+0.025, +0.046]
Same, TF-IDF, all train			+0.028 [+0.021, +0.037]

FAR: false-alarm rate on ham. Single-message rules: the header and request verifiers alone. Learned systems are held to the routed score's validation false-alarm rate with a 0.5% floor. Intervals: 1,000 resamples of whole subject groups or threads.

A. Pooled Detection

Table II(a) compares the routed score with four learned systems that use the same information. On 3,045 test attacks against 6,357 test ham it detects fewer attacks than any of them: its F1 is 0.238 [0.204, 0.277] and its recall 13.6%, against an F1 of 0.639 for text only, 0.788 for headers only, 0.882 for parallel fusion and 0.658 for the flat classifier. All four paired differences in F1 lie below 0; routed minus fusion is −0.644 [−0.678, −0.605]. The AUC, which needs no cut, is also the lowest (0.812 against 0.949 to 0.993). Three things limit this comparison and none removes the gap: the learned systems were fitted on attack labels and the routed score uses none, a headers-only model can partly learn which corpus an email came from, and the 0.5% floor lets each learned system flag more validation ham than the routed score does. The attack sources also differ in their header evidence: headers only detects 99.5% of the phishing_pot attacks but 2.5% of the Nigerian fraud emails, whose original headers were not kept, while the routed score detects 7.4%, 15.3% and 22.8% of the phishing_pot, Nazario and Nigerian fraud attacks.

Because only contradictions can lift an email to Suspicious, the routed score can flag only emails with a checked claim, and 788 of the 3,045 test attacks have one (398 of the 6,357 ham). On these emails its F1 is 0.679 [0.633, 0.734]. It is not distinguishable from text only (difference −0.063 [−0.136, +0.026]) or headers only (−0.057 [−0.119, +0.003]), remains behind parallel fusion (−0.260 [−0.305, −0.211]) and the flat classifier (−0.174 [−0.214, −0.135]), and has a higher false-alarm rate on that ham than text only or headers only (4.3% against 1.5% and 0.5%).

Every email the routed score flagged has a claim, a rule and evidence behind it (442 of 442); none of the flat classifier's 1,819 flags names a claim. This is a property of the design, not a measured gain. On the 31 synthetic colleague-impersonation pairs the routed score flagged 19 attacks (61.3% [45.2%, 77.4%]) and 0 twins. Headers only flagged all 31 attacks and all 31 twins, parallel fusion 17 attacks and 4 twins, the flat classifier 17 and 0, and text only 11 and 0. The text and the headers of this arm are synthetic.

B. Thread Consistency

On variant A, which changes only the content, the thread rules detect 18 of 25 Apache cases (72.0% [52.0%, 88.0%]) and 48 of 51 Enron cases (94.1% [88.2%, 100.0%]), against 5 of 25 and 0 of 51 for the header and request rules alone (Table II(b)). Variants B and C each change one thing the rules are defined to check, so their detection is close to a check that the rules do what they are defined to do. When the whole thread is scanned, the first contradiction falls on the injected message in 17 of 25 Apache and 40 of 51 Enron takeover cases. The thread rules flagged 0 of 25 real next replies on Apache and 1 of 51 on Enron. On all real, unmodified test threads they raised a medium or high contradiction on 4.44% of 698 Enron messages with earlier messages [3.08%, 5.99%], 0.50% of 201 Tomcat messages and 0.00% of 95 Kafka messages. The injected messages and their headers are synthetic, so these rates test the rules against the construction, not against real attackers.

C. Payload-Free Protocol

With corpus labels and one training sample of 9,856 emails and two seeds, model A loses attack-class F1 when links are redacted at test time (raw minus redacted +0.059 [+0.050, +0.069]); model B does not (−0.014 [−0.019, −0.009]). A's drop minus B's drop is above 0 in all three families (Table II(c)), so part of a raw-trained detector's accuracy is link-reading. Training on redacted bodies keeps most of the accuracy: B on redacted text scores 0.932 against 0.935 for A on raw text. On the 696 naturally link-free test attacks the two DistilBERT models are not distinguishable (0.863 [0.840, 0.884] and 0.862 [0.839, 0.884]), but that subset is a different mix of sources, so its F1 is not a before-and-after of the same emails. A link-presence rule alone reaches an F1 of 0.413 at a ham false-positive rate of 46.5%.

D. Tactics, Claims and Score Budget

On the 140 labelled test emails, whose labels come from one LLM family, the tactic classifier's macro-F1 over authority, urgency, scarcity and secrecy is 0.479 [0.370, 0.574], against 0.500 [0.408, 0.581] for a keyword baseline with thresholds tuned on validation; the paired difference, −0.021 [−0.144, +0.091], includes 0, so neither is better. The claim extractor's macro-F1 over the five claim types with at least 10 positives is 0.568 [0.470, 0.646]; its weakest type is internal affiliation (F1 0.250 [0.000, 0.533], 2 of 12 found), on which the two annotators agree with each other at an F1 of only 0.410. The frozen score stayed within its false-alarm budget (High risk for at most 1% and Suspicious or above for at most 5% of checked legitimate mail per source) in 4 of 5 groups with at least 20 checked emails. CEAS-08 ham was over, at 5.60% of 268 checked emails, with an interval [3.42%, 9.03%] that still includes the budget.

VII. Discussion and Limitations

The results support the payload-free protocol, the thread checks and the traceability of alarms, within the limits below. They do not support the expectation that claim-conditioned checking detects more attacks than simpler uses of the same information: on these corpora it detects fewer.

Several limits apply. Tactic and claim scores are agreement with LLM labels from one model family; the N1 labels are corpus labels. No source holds both attacks and ordinary mail, so every pooled number mixes corpus and detection, and the collections differ by wording alone: real and synthetic attack emails are separated by their words with an AUC of 1.000 [1.000, 1.000] (90 real and 31 synthetic test emails), which is why they are reported apart. The learned comparators saw attack labels, and a headers-only model can partly learn which corpus an email came from; both favour them. The hijack cases, the synthetic header blocks and the paraphrases are machine-written: they show that the rules do what they are defined to do, not how often real attackers behave that way. N1 rests on one training sample and two seeds, so its intervals cover the test sample and not the training randomness. Rewording 144 attack bodies with a language model kept 18 of the 24 flags they started with, but the headers were untouched, so a header contradiction survives by construction, and meaning was checked on six pairs only. Internal-affiliation checks need the organisation's domain, which most public corpora do not reveal, and the extractor reads English only. An attacker who copies the identifiers, the sending details and the tone of a real thread is not caught by any rule here.

VIII. Conclusion

Checking what an email claims against its own headers and thread gives every alarm a stated reason and finds synthetic takeover replies in real threads that single-message rules miss. On pooled detection over public corpora, however, the routed score is behind text-only, header-only, fused and flat classifiers, so the design does not by itself detect more attacks on these data. Redacting links is cheap and removes a shortcut. Testing the approach on real attacks needs labelled payload-free BEC and real hijacked threads that keep receiving-side authentication headers, and a human-labelled sample for tactics and claims.

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